PROJECT REPORT

Heart Disease Risk Factor Analysis

Using Tableau Dashboards and Flask Web Integration

Submitted To	SmartBridge Program Authorities
Department	Computer Science and Information Technology
Technology Stack	Tableau, Flask (Python), Render
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Status	Final Submission

Executive Summary

This report presents a comprehensive analysis of heart disease risk factors through interactive data visualization. The project leverages Tableau dashboards integrated with a Flask web application to enable healthcare professionals, researchers, and policy makers to explore population health patterns and identify high-risk groups for preventive action.

Key Highlights

- Interactive Tableau dashboards covering demographic, lifestyle, and clinical risk factors
- Flask-based web application deployed on Render Cloud Platform
- Three integrated dashboards with real-time filtering and responsive visualization
- Average dashboard load performance: 2–4 seconds
- Publicly accessible via cloud deployment for healthcare stakeholders

1. Project Overview

1.1 Problem Domain

Heart disease is one of the leading causes of death worldwide. Understanding lifestyle, demographic, and medical risk factors is critical for prevention and treatment planning. Traditional analysis tools lack the interactivity and accessibility needed for real-time healthcare decision support.

1.2 System Overview

This project addresses these challenges by developing an end-to-end analytics platform. Tableau is used to build interactive dashboards and Flask (Python) serves as the web framework to embed and deploy them through a publicly accessible web interface hosted on Render Cloud.

1.3 Core Technology Stack

Component	Technology
Visualization	Tableau Desktop / Tableau Public

Backend Framework	Flask (Python 3.x)
Frontend	HTML5, CSS3, JavaScript
WSGI Server	Gunicorn
Version Control	Git / GitHub
Deployment Platform	Render Cloud Platform

1.4 Target Users & Real-World Relevance

The platform is designed to serve multiple stakeholders across the healthcare ecosystem:

- Healthcare Professionals — Analyze patient data to identify major risk factors such as smoking, obesity, and diabetes.
- Government Health Departments — Analyze population health trends and design public awareness programs.
- Researchers & Academics — Use real healthcare datasets to study risk patterns and outcomes.
- Students — Learn data analytics, visualization, and dashboard design in a practical context.
- Individuals — Visualize lifestyle risks and understand preventive health measures.

2. Technical Architecture

2.1 System Architecture Flow

```
User Browser → Flask Web Application → Tableau Dashboards → Dataset
```

2.2 Layer Architecture

Layer	Details
Frontend Layer	HTML, CSS, JavaScript — User interface and dashboard embedding
Backend Layer	Flask (Python) — Server-side routing and application logic
Visualization Layer	Tableau Dashboards — Interactive charts and analytical views
Deployment Layer	Render Cloud Platform — Public hosting and service delivery

3. System Requirements

3.1 Software Requirements

Requirement	Details
Python	Version 3.x or higher
Tableau	Tableau Desktop or Tableau Public
IDE	VS Code or any compatible IDE
Version Control	Git
Browser	Any modern web browser
Libraries	Flask, Gunicorn (via requirements.txt)

3.2 Hardware Requirements

Component	Specification
Processor	Intel i3 or higher
RAM	Minimum 4 GB
Storage	Minimum 2 GB free space

3.3 Prior Knowledge Required

- Basic Python programming
- Flask web framework fundamentals
- Data visualization concepts
- Tableau dashboard creation
- Basic HTML/CSS
- Git and GitHub fundamentals

4. Project Objectives

Objective Category	Description
Technical	Develop interactive Tableau dashboards and integrate them with Flask web framework.

Performance	Ensure smooth dashboard interaction with average load time of 2–4 seconds.
Deployment	Host the complete web application on Render for public cloud access.
Learning Outcomes	Gain practical experience in data analytics, visualization, and web deployment.

5. Project Milestones

Milestone 1 — Requirement Analysis & System Design

Problem Definition: Analyze heart disease risk factors using modern visualization techniques.

Functional Requirements:

- Display interactive dashboards
- Visualize health trends across demographics
- Provide dynamic filtering capabilities

Non-Functional Requirements:

- Fast loading dashboards
- User-friendly interface
- Cloud deployment capability

Milestone 2 — Environment Setup & Configuration

Development environment established with Python, Flask, Gunicorn, and Tableau Desktop. Project folder structure initialized as follows:

```
heart_disease_dashboard/
├── app.py
├── requirements.txt
├── templates/
│   └── dashboards.html
└── static/  (css, js)
```

Milestone 3 — Core System Development

Feature	Description

Data Visualization	Gender vs Heart Disease, Age vs Heart Disease, Smoking & Alcohol Impact, Physical Activity vs Heart Disease
Dashboard Design	Three dashboards: (1) Demographic & Lifestyle Factors, (2) Health Conditions & Population Insights, (3) BMI, Diabetes, and Stroke Risk
Web Integration	All dashboards embedded into Flask web application using Tableau embed code

Milestone 4 — Integration & Optimization

- Integrated Flask backend with Tableau dashboards
- Optimized layout for responsive viewing across devices
- Handled embedding and UI adjustments for seamless user experience

Milestone 5 — Testing & Validation

Test Case	Description
TC01	Website loads successfully
TC02	Dashboards render correctly on page load
TC03	Filters update visualizations in real-time
TC04	Dashboard opens in fullscreen mode

Performance Metric: Average dashboard load time: 2–4 seconds

6. Deployment

Deployment Platform: Render Cloud Hosting

Deployment was executed through the following sequence:

1. Upload project repository to GitHub.
2. Connect GitHub repository to Render platform.
3. Configure build command and start command in Render settings.
4. Deploy the Flask application to public cloud endpoint.

Live Tableau dashboards are embedded from Tableau Public and served through the Render-hosted Flask application.

7. Results & Outcomes

- System successfully visualizes heart disease risk factors across multiple dimensions.
- Users can explore healthcare insights through fully interactive dashboards.
- Platform effectively demonstrates how visualization tools reveal patterns in healthcare data.
- Application is publicly accessible via cloud deployment.

8. Advantages & Limitations

8.1 Advantages

- Interactive analytics enabling real-time data exploration
- Easy-to-understand visual dashboards for non-technical users
- Web-based accessibility — no software installation required for end users

8.2 Limitations

- Depends on Tableau Public hosting for dashboard availability
- Large dashboards may increase loading time on low-bandwidth connections

9. Future Enhancements

- Machine learning based risk prediction model integration
- Mobile-optimized dashboard interface
- Real-time healthcare data integration via APIs
- AI-based personalized health recommendation system

10. Conclusion

This project demonstrates the power of data visualization in healthcare analytics. By integrating Tableau dashboards with a Flask web application, users can interactively explore heart disease risk factors across demographic, lifestyle, and clinical dimensions.

The system highlights the importance of lifestyle and medical indicators in understanding heart disease risk and provides a scalable analytical tool for researchers, healthcare professionals, and policy makers. The successful deployment on Render Cloud Platform ensures broad accessibility and demonstrates the viability of cloud-based healthcare analytics solutions.

Appendix

Item	Details
Dataset	https://drive.google.com/file/d/190Qmq27LeZZ_nWricP3Obl7ys_5otEsp/view
GitHub Repository	https://github.com/itsabhinavdevs/heart_disease_dashboard_analysis_tableau
Live Demo URL	https://heart-disease-dashboard-analysis-ta.vercel.app/